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(54) **RESISTIVE ATTENUATOR AND METHOD FOR IMPROVING LINEARITY OF RESISTIVE ATTENUATOR**

(71) Applicant: **Realtek Semiconductor Corp.,**
HsinChu (TW)

(72) Inventors: **Yi-Ching Wu, HsinChu (TW);**
Chia-Jun Chang, HsinChu (TW)

(73) Assignee: **Realtek Semiconductor Corp.,**
HsinChu (TW)

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H03H 7/25 (2006.01)
H03H 11/46 (2006.01)
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CPC **H03H 11/53** (2013.01); **H03F 1/3282**
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See application file for complete search history.

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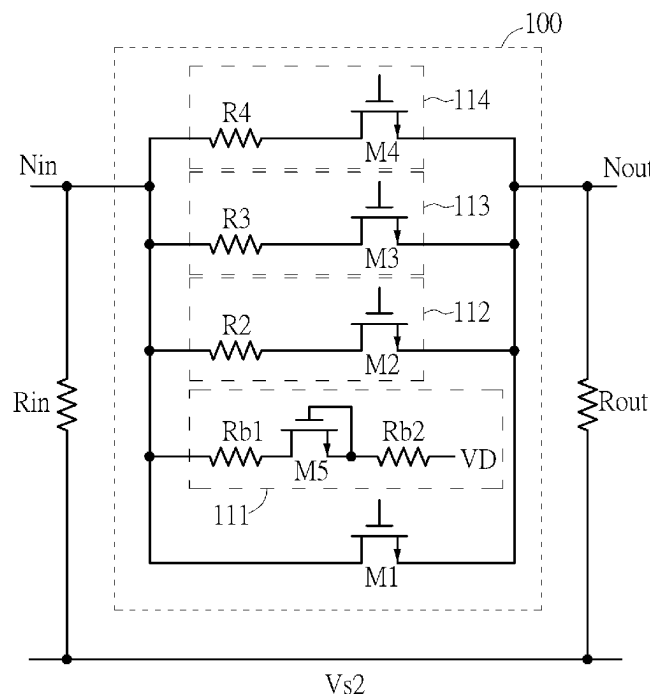
Primary Examiner — Tomi Skibinski

(74) Attorney, Agent, or Firm — Winston Hsu

(57) **ABSTRACT**

A resistive attenuator and a method for improving linearity of the resistive attenuator are provided. The resistive attenuator includes a first transistor, an attenuation circuit and a compensation circuit, wherein both the first transistor and the attenuation circuit are coupled between an input terminal and an output terminal of the resistive attenuator, and the compensation circuit is coupled to the first transistor. The first transistor is configured to provide a first signal path between the input terminal and the output terminal. The attenuation circuit is configured to provide a second signal path between the input terminal and the output terminal, wherein signal attenuation of the second signal path is greater than signal attenuation of the first signal path. The compensation circuit is configured to compensate nonlinear distortion caused by the first transistor.

8 Claims, 3 Drawing Sheets



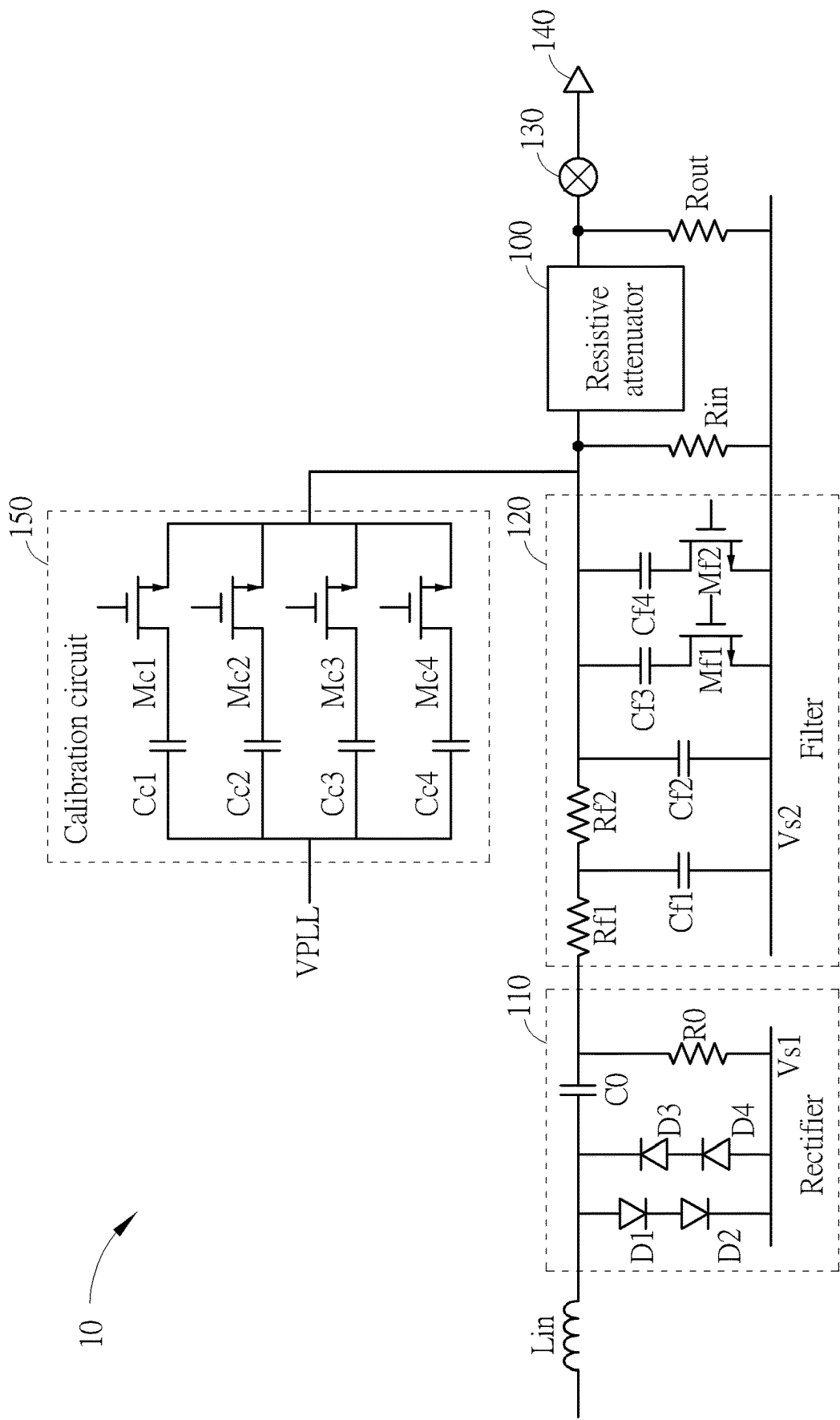


FIG. 1

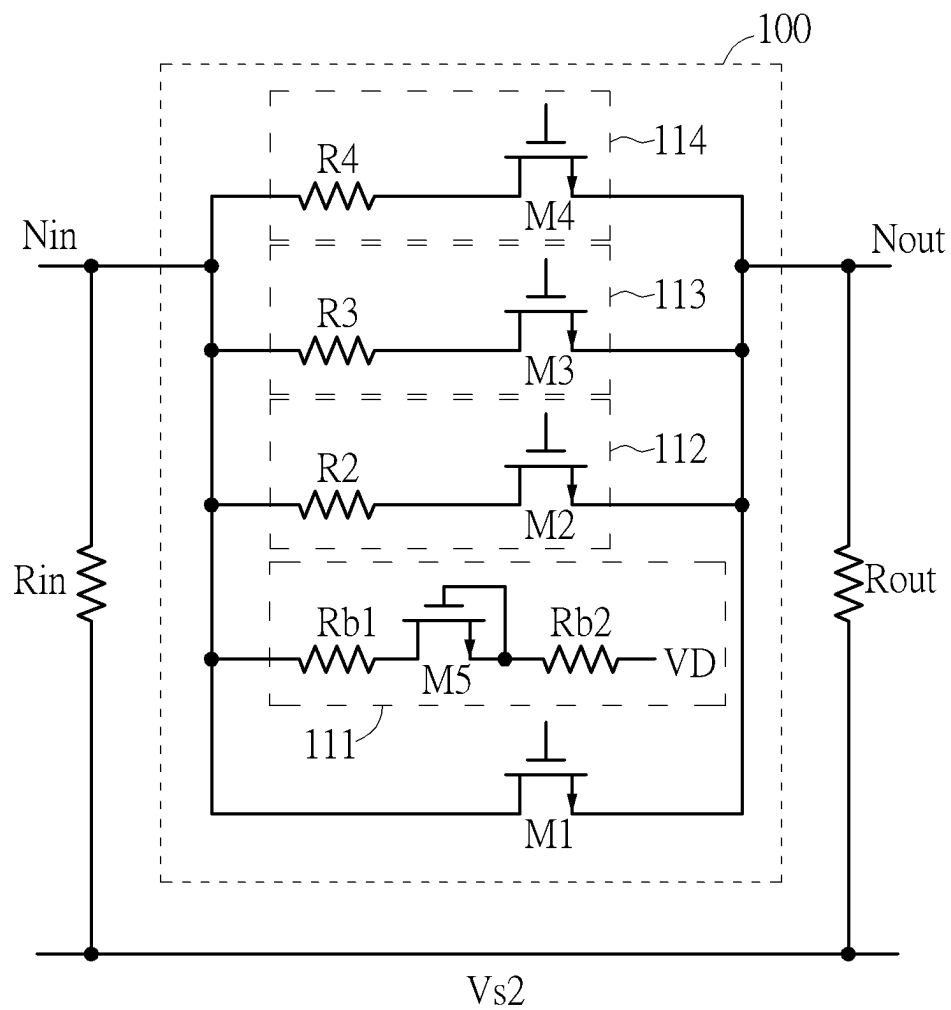


FIG. 2

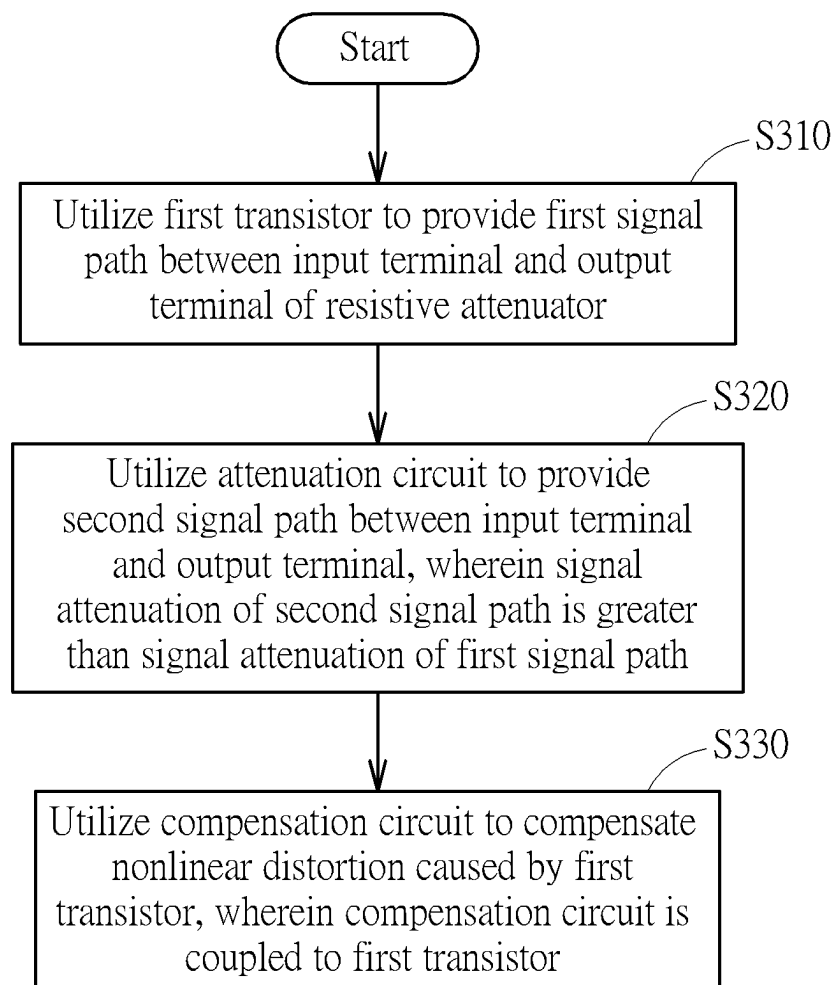


FIG. 3

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RESISTIVE ATTENUATOR AND METHOD FOR IMPROVING LINEARITY OF RESISTIVE ATTENUATOR

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention is related to attenuator circuits, and more particularly, to a resistive attenuator and a method for improving linearity of the resistive attenuator.

2. Description of the Prior Art

As an external digital pre-distortion circuit typically operates under a condition where a signal power is 0 decibels relative to one milliwatt (dBm), design optimization is performed based on this scenario. In some situations, the external digital pre-distortion circuit needs to handle signals of higher power such as 10 dBm. When the external digital pre-distortion circuit designed based on the 0 dBm signal power condition handles a signal of 10 dBm power, overall linearity may not achieve a target specification: for example, performance related to amplitude-to-amplitude (AMAM) distortion and amplitude-to-phase (AMPM) distortion may fail to meet requirements. The above linearity issues are typically caused by an attenuator within the external digital pre-distortion circuit.

Some related arts solutions add additional loss on signal paths in order to guarantee that the external digital pre-distortion circuit still achieves good linearity while handling the 10 dBm power signal. This solution, however, will mean a corresponding noise figure fails to meet requirements when the external digital pre-distortion circuit handles 0 dBm power signals.

Thus, there is a need for a novel architecture and an associated method, in order to improve linearity of the external digital pre-distortion circuit (and more particularly, the attenuator therein) without introducing any side effect or in a way that is less likely to introduce side effects.

SUMMARY OF THE INVENTION

An objective of the present invention is to provide a resistive attenuator and a method for improving linearity of the resistive attenuator which can make the resistive attenuator able to improve linearity when handling high power signal without affecting performance related to the noise figure or reducing the effect on the noise figure related performance.

At least one embodiment of the present invention provides a resistive attenuator. The resistive attenuator comprises a first transistor, at least one attenuation circuit and a compensation circuit, wherein the first transistor is coupled between an input terminal and an output terminal of the resistive attenuator, the at least one attenuation circuit is coupled between the input terminal and the output terminal, and the compensation circuit is coupled to the first transistor. More particularly, the first transistor is configured to provide a first signal path between the input terminal and the output terminal. The at least one attenuation circuit is configured to provide at least one second signal path between the input terminal and the output terminal, wherein signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path. In addition, the compensation circuit is configured to compensate nonlinear distortion caused by the first transistor.

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At least one embodiment of the present invention provides a method for improving linearity of a resistive attenuator. The method comprises: utilizing a first transistor of the resistive attenuator to provide a first signal path between an input terminal and an output terminal of the resistive attenuator; utilizing at least one attenuation circuit of the resistive attenuator to provide at least one second signal path between the input terminal and the output terminal, wherein signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path; and utilizing a compensation circuit of the resistive attenuator to compensate nonlinear distortion caused by the first transistor, wherein the compensation circuit is coupled to the first transistor.

The resistive attenuator and the method provided by the embodiment of the present invention utilize configuration of the compensation circuit to cancel nonlinear distortion caused by the transistor coupled between the input terminal and the output terminal in the resistive attenuator. As the compensation circuit will not introduce additional loss, linearity of the resistive attenuator can be improved without sacrificing performance related to the noise figure. In addition, the embodiments of the present invention will not greatly increase additional costs. Thus, the present invention can solve the problem of the related art without introducing any side effect or in a way that is less likely to introduce side effects.

These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a digital pre-distortion circuit according to an embodiment of the present invention.

FIG. 2 is a diagram illustrating details of the digital pre-distortion circuit shown in FIG. 1 according to an embodiment of the present invention.

FIG. 3 is a diagram illustrating a working flow of a method for performing linearity of a resistive attenuator according to an embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 is a diagram illustrating a digital pre-distortion (DPD) circuit such as an external DPD (EDPD) circuit 10 according to an embodiment of the present invention, where the EDPD circuit 10 at least comprises a resistive attenuator 100. In this embodiment, the EDPD circuit may further comprise an input inductor Lin, a rectifier 110, a filter 120, an input bias resistor Rin, an output bias resistor Rout, a mixer 130, a transimpedance amplifier (TIA) 140 and a calibration circuit 150. More particularly, the EDPD circuit 10 may receive an input signal via the input inductor Lin. After being processed by the rectifier 110 and the filter 120, the resistive attenuator 100 may perform attenuation upon a signal output from the filter 120, to allow the mixer 130 and the TIA 140 generate an output signal according to an attenuated signal output from the resistive attenuator 100. In addition, the calibration circuit 150 may be configured to perform image rejection ratio calibration of the EDPD 10, where the calibration circuit 150 may receive a signal VPLL from a phase locked loop (PLL), and accordingly generate

a corresponding test signal for simulating a signal from a signal path of the input inductor L_{in} , the rectifier **110** and the filter **120**.

As shown in FIG. 1, the rectifier **110** may comprise diodes $D1$, $D2$, $D3$ and $D4$, a capacitor $C0$ and a resistor $R0$, where the rectifier **110** operates by taking a voltage level of a reference voltage $Vs1$ as a reference level. The filter **120** may comprise resistors $Rf1$ and $Rf2$, capacitors $Cf1$, $Cf2$, $Cf3$ and $Cf4$, and transistors $Mf1$ and $Mf2$, where the filter **120** operates by taking a voltage level of a reference voltage $Vs2$ as a reference level, and each of the transistors $Mf1$ and $Mf2$ may serve as a switch. The calibration circuit **150** may comprise capacitors $Cc1$, $Cc2$, $Cc3$ and $Cc4$, and transistors $Mc1$, $Mc2$, $Mc3$ and $Mc4$, where each of the capacitors $Cc1$, $Cc2$, $Cc3$ and $Cc4$ may perform alternating current (AC) coupling upon the signal V_{PLL} to extract an AC component within the signal V_{PLL} , and each of the transistors $Mc1$, $Mc2$, $Mc3$ and $Mc4$ may serve as a switch. It should be noted that the present invention focuses on implementation of the resistive attenuator **100**, where the other peripheral components are main points of the present invention, and detailed operations of these peripheral components are omitted here for brevity. In addition, the above implementation details are examples of these peripheral components, but the present invention is not limited thereto.

FIG. 2 is a diagram illustrating details of the digital pre-distortion circuit shown in FIG. 1 according to an embodiment of the present invention. As shown in FIG. 2, the resistive attenuator **100** may comprise a transistor $M1$, at least one attenuation circuit such as attenuation circuits **112**, **113** and **114**, and a compensation circuit **111**, where the transistor $M1$ is coupled to an input terminal N_{in} and an output terminal N_{out} of the resistive attenuator **100**, each of the attenuation circuits **112**, **113** and **114** are coupled between the input terminal N_{in} and the output terminal N_{out} , and the compensation circuit **111** is coupled to the transistor $M1$ (e.g. coupled between a reference voltage V_D and the input terminal N_{in}). In this embodiment, the transistor $M1$ may be configured to provide a first signal path between the input terminal N_{in} and the output terminal N_{out} . Each of the attenuation circuits **112**, **113** and **114** may be configured to provide at least one second signal path between the input terminal N_{in} and the output terminal N_{out} , where signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path. For better illustration, the first signal path may be referred to as a low loss path, and the at least one second signal path may be referred to as a high loss path. In addition, the compensation circuit **111** may be configured to compensate nonlinear distortion caused by the transistor $M1$.

In this embodiment, each of the attenuation circuits **112**, **113** and **114** may comprise an attenuation resistor and a switch transistor, and the switch transistor and the attenuation resistor are connected in series between the input terminal N_{in} and the output terminal N_{out} . In detail, the attenuation circuit **112** may comprise an attenuation resistor $R2$ and a switch transistor $M2$, and the switch transistor $M2$ and the attenuation resistor $R2$ are connected in series between the input terminal N_{in} and the output terminal N_{out} . The attenuation circuit **113** may comprise an attenuation resistor $R3$ and a switch transistor $M3$, and the switch transistor $M3$ and the attenuation resistor $R3$ are connected in series between the input terminal N_{in} and the output terminal N_{out} . The attenuation circuit **114** may comprise an attenuation resistor $R4$ and a switch transistor $M4$, and the switch transistor $M4$ and the attenuation resistor $R4$ are

connected in series between the input terminal N_{in} and the output terminal N_{out} . In addition, a gate terminal of the transistor $M1$ is configured to control whether to enable the low loss path provided by the transistor $M1$, a gate terminal of the transistor $M2$ is configured to control whether to enable the high loss path provided by the attenuation circuit **112**, a gate terminal of the transistor $M3$ is configured to control whether to enable the high loss path provided by the attenuation circuit **113**, and a gate terminal of the transistor $M4$ is configured to control whether to enable the high loss path provided by the attenuation circuit **114**.

In this embodiment, overall signal attenuation between the input terminal N_{in} and the output terminal N_{out} may be determined according to whether any of the first signal path and the second signal path is enabled. For example, when all of the transistors $M1$, $M2$, $M3$ and $M4$ are turned on, the resistive attenuator **100** may provide a first attenuation rate between the input terminal N_{in} and the output terminal N_{out} . When the transistor $M1$ is turned off and the transistors $M2$, $M3$ and $M4$ are turned on, the resistive attenuator **100** may provide a second attenuation rate between the input terminal N_{in} and the output terminal N_{out} . When the transistors $M1$ and $M2$ are turned off and the transistors $M3$ and $M4$ are turned on, the resistive attenuator **100** may provide a third attenuation rate between the input terminal N_{in} and the output terminal N_{out} . When the transistors $M1$, $M2$ and $M3$ are turned off and the transistor $M4$ is turned on, the resistive attenuator **100** may provide a fourth attenuation rate between the input terminal N_{in} and the output terminal N_{out} .

As the low loss path provided by the transistor $M1$ does not have a transistor connected in series with the transistor $M1$, however, when the transistor $M1$ is turned on, a signal on the input terminal N_{in} may have a high power transmitted to the output terminal N_{out} through the transistor $M1$, making a signal conversion between the input terminal N_{in} and the output terminal N_{out} have a lot of nonlinear components (e.g. a third order term). In order to prevent linearity of the resistive attenuator **100** from getting worse due to the transistor $M1$, the present invention utilizes the compensation circuit **111** to generate a reverse third order term to cancel the third order term generated by the transistor $M1$ for compensating the nonlinear distortion caused by the transistor $M1$.

As shown in FIG. 2, the compensation circuit **111** may comprise the transistor $M5$, where a gate terminal of the transistor $M5$ is coupled to a source terminal of the transistor $M5$. In addition, the compensation circuit **111** may further comprise bias resistors $Rb1$ and $Rb2$, where the bias resistor $Rb1$ is coupled between the transistors $M5$ (e.g. coupled to a drain terminal of the transistor $M5$) and the transistor $M1$, and the bias resistor $Rb2$ is coupled between the transistor $M5$ (e.g. coupled to the source terminal of the transistor $M5$) and the reference voltage V_D . In particular, the bias resistors $Rb1$ and $Rb2$ may be configured to control a drain-to-source voltage difference of the transistor $M5$, to make nonlinear distortion caused by the transistor $M5$ and the nonlinear distortion caused by the transistor $M1$ be canceled by each other. In some embodiments, the transistors $M5$ and $M1$ may have the same size (e.g. channel width and channel length). In other embodiments, the transistors $M5$ and $M1$ may have different sizes. In some embodiments, the bias resistors $Rb1$ and $Rb2$ may have the same resistance. In other embodiments, the bias resistors $Rb1$ and $Rb2$ may have different resistances. As long as the transistor $M5$ under bias control of the bias resistors $Rb1$ and $Rb2$ generates a third order term having the same (or similar) magnitude and opposite

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direction as the transistor M1, designs of the size of the transistor M5 and the resistances of the bias resistors Rb1 and Rb2 may vary.

TABLE 1

Config- uration	AMAM (Enable compensation circuit)	AMAM (Disable compensation circuit)	AMPM (Enable compensation circuit)	AMPM (Disable compensation circuit)
#1	-0.279	-0.432	-0.932	-1.07
#2	-0.151	-0.252	-0.636	-0.882
#3	-0.063	-0.116	-0.364	-0.538
#4	-0.013	-0.024	-0.183	-0.232

Table 1 shows amplitude-to-amplitude (AMAM) distortion and amplitude-to-phase (AMPM) distortion for illustrating linearity improvements obtained from the compensation circuit 111 of the present invention. The second column to the fifth column of Table 1 represent, respectively, AMAM distortion (with values in decibels) of the resistive attenuator 100 under a condition where the compensation circuit 111 is enabled, AMAM distortion (with values in decibels) of the resistive attenuator 100 under a condition where the compensation circuit 111 is disabled, AMPM distortion (with values in degrees) of the resistive attenuator 100 under the condition where the compensation circuit 111 is enabled, and AMPM distortion (with values in degrees) of the resistive attenuator 100 under the condition where the compensation circuit 111 is disabled. The second row to the fifth row of Table 1 represent, respectively, the resistive attenuator 100 configured to have the first attenuation rate, the second attenuation rate, the third attenuation rate and the fourth attenuation rate. For example, configuration #1 shown in Table 1 represents that all of the transistors M1, M2, M3 and M4 are turned on, configuration #2 shown in Table 1 represents that the transistor M1 is turned off and the transistors M2, M3 and M4 are turned on, configuration #3 shown in Table 1 represents that the transistors M1 and M2 are turned off and the transistors M3 and M4 are turned on, and configuration #4 shown in Table 1 represents that the transistors M1, M2 and M3 are turned off and the transistor M4 is turned on. As shown in Table 1, in comparison with the condition of turning off the compensation circuit 111, the AMAM distortion and the AMPM distortion of the resistive attenuator 100 can be effectively reduced when the compensation circuit 111 is enabled, where lower absolute values of the AMAM distortion and the AMPM distortion shown in Table 1 means better linearity. It should be noted that there are still some parasitic capacitors (e.g. gate-to-source capacitors or gate-to-drain capacitors) under a condition where the transistor M1 is turned off, making the transistor M1 (which is turned off) continue to generate nonlinear terms which impact the linearity of the resistive attenuator 100. Thus, even though the transistor M1 is turned off when the resistive attenuator 100 is configured to have any of the second attenuation rate, the third attenuation rate, and the fourth attenuation rate, the linearity of the resistive attenuator 100 still can be improved because of using the compensation circuit 111.

In addition the resistive attenuator 100 can obtain improvements in performance related to third-order-intermodulation distortion (IMD3) by using the compensation circuit 111. In particular, regardless of testing with an input signal of 10 decibels relative to one milliwatt (dBm) or 5 dBm, IMDs results can be improved when the resistive attenuator 100 is configured to have any of the first attenu-

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ation rate, the second attenuation rate, the third attenuation rate and the fourth attenuation rate because of using the compensation circuit 111, and more particularly, results of both upper band IMD3 and lower band IMD3 can be improved.

FIG. 3 is a diagram illustrating a working flow of a method for performing linearity of a resistive attenuator (e.g. the resistive attenuator 100 shown in FIG. 2) according to an embodiment of the present invention. It should be noted that the working flow shown in FIG. 3 is for illustrative purposes only, and is not meant to be a limitation of the present invention. More particularly, if a same result can be obtained, one or more steps may be added, deleted or modified in the working flow shown in FIG. 3. In addition, these steps do not have to be executed in the exact order shown in FIG. 3.

In Step S310, the resistive attenuator may utilize a first transistor therein to provide a first signal path between an input terminal and an output terminal of the resistive attenuator.

In Step S320, the resistive attenuator may utilize at least one attenuation circuit therein to provide at least one second signal path between the input terminal and the output terminal, wherein signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path.

In Step S330, the resistive attenuator may utilize a compensation circuit therein to compensate nonlinear distortion caused by the first transistor, wherein the compensation circuit is coupled to the first transistor.

To summarize, the embodiments of the present invention can make a nonlinear term generated by the transistor M5 and a nonlinear term generated by the transistor M1 cancel each other by controlling bias of the transistor M5 within the compensation circuit 111. Thus, the embodiments of the present invention can reduce nonlinear distortion caused by the transistor M1 as much as possible without connecting additional resistor(s) in series with the transistor M1. In addition, the embodiments of the present invention will not greatly increase additional costs. Thus, the present invention can solve the problem of the related art without introducing any side effect or in a way that is less likely to introduce side effects.

Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A resistive attenuator, comprising:

a first transistor, coupled between an input terminal and an output terminal of the resistive attenuator, configured to provide a first signal path between the input terminal and the output terminal;

at least one attenuation circuit, coupled between the input terminal and the output terminal, configured to provide at least one second signal path between the input terminal and the output terminal, wherein signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path; and

a compensation circuit, coupled to the first transistor, configured to compensate nonlinear distortion caused by the first transistor;

wherein the compensation circuit comprises a second transistor, and a gate terminal of the second transistor is coupled to a source terminal of the second transistor.

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2. The resistive attenuator of claim 1, wherein the compensation circuit further comprises:

a first bias resistor, coupled between the first transistor and the second transistor; and

a second bias resistor, coupled between the second transistor and a reference voltage;

wherein the first bias resistor and the second bias resistor are configured to control a drain-to-source voltage difference of the second transistor, to make nonlinear distortion caused by the second transistor cancel and the nonlinear distortion caused by the first transistor cancel each other.

3. The resistive attenuator of claim 1, wherein the at least one attenuation circuit comprises an attenuation resistor and a switch transistor, and the switch transistor and the attenuation resistor are connected in series between the input terminal and the output terminal.

4. The resistive attenuator of claim 3, wherein a gate terminal of the first transistor is configured to control whether to enable the first signal path, a gate terminal of the switch transistor is configured to control whether to enable the at least one second signal path, and overall signal attenuation between the input terminal and the output terminal is determined according to whether any of the first signal path and the at least one second signal path is enabled.

5. A method for improving linearity of a resistive attenuator, comprising:

utilizing a first transistor of the resistive attenuator to provide a first signal path between an input terminal and an output terminal of the resistive attenuator;

utilizing at least one attenuation circuit of the resistive attenuator to provide at least one second signal path between the input terminal and the output terminal, wherein signal attenuation of the at least one second signal path is greater than signal attenuation of the first signal path; and

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utilizing a compensation circuit of the resistive attenuator to compensate nonlinear distortion caused by the first transistor, wherein the compensation circuit is coupled to the first transistor;

wherein the compensation circuit comprises a second transistor, and a gate terminal of the second transistor is coupled to a source terminal of the second transistor.

6. The method of claim 5, wherein the compensation circuit further comprises a first bias resistor and a second bias resistor, the first bias resistor is coupled between the first transistor and the second transistor, the second bias resistor is coupled between the second transistor and a reference voltage, and utilizing the compensation circuit of the resistive attenuator to compensate the nonlinear distortion caused by the first transistor comprises:

utilizing the first bias resistor and the second bias resistor to control a drain-to-source voltage difference of the second transistor, to make nonlinear distortion caused by the second transistor cancel and the nonlinear distortion caused by the first transistor cancel each other.

7. The method of claim 5, wherein the at least one attenuation circuit comprises an attenuation resistor and a switch transistor, and the switch transistor and the attenuation resistor are connected in series between the input terminal and the output terminal.

8. The method of claim 7, further comprising:

utilizing a gate terminal of the first transistor to control whether to enable the first signal path; and

utilizing a gate terminal of the switch transistor to control whether to enable the at least one second signal path; wherein overall signal attenuation between the input terminal and the output terminal is determined according to whether any of the first signal path and the at least one second signal path is enabled.

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